ECET 250E-Linear Circuit 1
Problemset 3-Chapter 3- Methods of Analysis
Solution

Name: _____ Marks: _____

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1. Chapter 3, Problem 1.

Given $\mathbf{R_1 = 4\ k\Omega}$, $\mathbf{R_2 = 2\ k\Omega}$, and $\mathbf{R_3 = 2\ k\Omega}$, determine the value of $\mathbf{I_x}$ using nodal analysis.

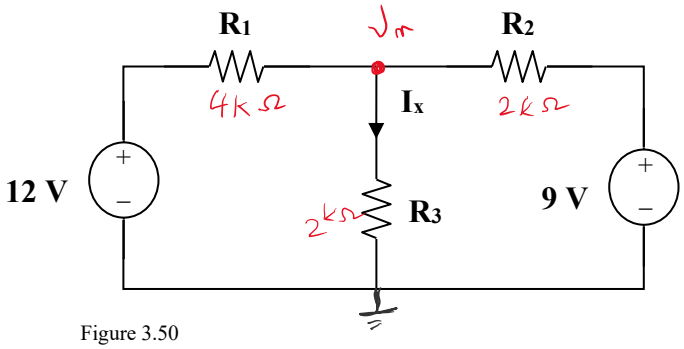


Figure 3.50

For Prob. 3.1

Chapter 3, Solution 1

Let the node voltage in the top middle of the circuit be designated as $\mathbf{V_x}$.

$[(V_x–12)/4k] + [(V_x–0)/2k] + [(V_x–9)/2k] = 0$ or (multiply this by 4 k)

$V_x – 12 + 2V_x + 2V_x – 18 = 0$

$(1+2+2)V_x = 12+18 = 30$ or $V_x = 30/5 = 6$ volts and

$I_x = 6/(2k) = \mathbf{3\ mA}$.

Answer: $I_x = 3$ mA

2. Chapter 3, Problem 2.

For the circuit in Fig. 3.51, obtain $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ using nodal analysis.

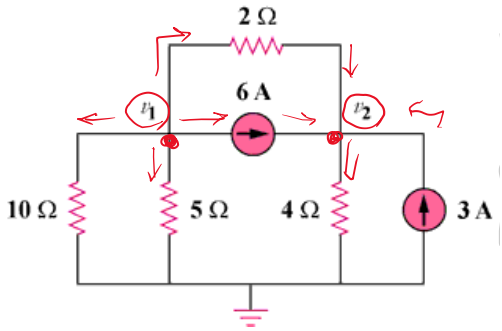


Figure 3.51

For Prob. 3.2.

Chapter 3, Solution 2

At node 1,

$$\frac{-v_1}{10} - \frac{v_1}{5} = 6 + \frac{v_1 - v_2}{2} \longrightarrow 60 = -8v_1 + 5v_2 \qquad (1)$$

At node 2,

$$\frac{v_2}{4} = 3 + 6 + \frac{v_1 - v_2}{2} \longrightarrow 36 = -2v_1 + 3v_2 \qquad (2)$$

Solving (1) and (2),

Answer: $v_1 = 0$ V, $v_2 = 12$ V

3. Chapter 3, Problem 3.

Find the currents i_1 through i_4 and the voltage v_o in the circuit in Fig. 3.52 using nodal analysis.

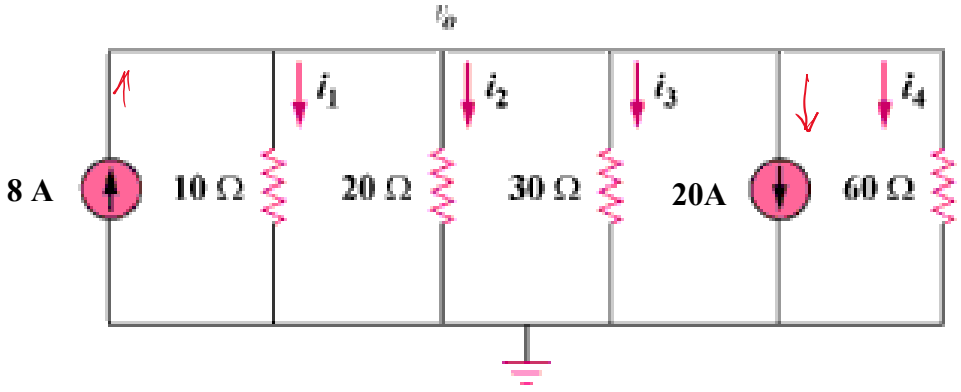


Figure 3.52
For Prob. 3.3.

Chapter 3, Solution 3

Applying KCL to the upper node,

$$-8 + \frac{v_0}{10} + \frac{v_o}{20} + \frac{v_o}{30} + 20 + \frac{v_0}{60} = 0 \text{ or } v_0 = \mathbf{-60\ V}$$

$$i_1 = \frac{v_0}{10} = \mathbf{-6\ A}, \; i_2 = \frac{v_0}{20} = \mathbf{-3\ A},$$

$$i_3 = \frac{v_0}{30} = -2\ \mathbf{A},\ i_4 = \frac{v_0}{60} = -1\ \mathbf{A}.$$

Answer: $v_0 = -60$ V; $i_1 = -6$ A, $i_2 = -3$ A, $i_3 = -2$ A, $i_4 = -1$ A

4. Chapter 3, Problem 4

Given the circuit in Fig. 3.53, calculate the currents i_1 through i_4 using nodal analysis.

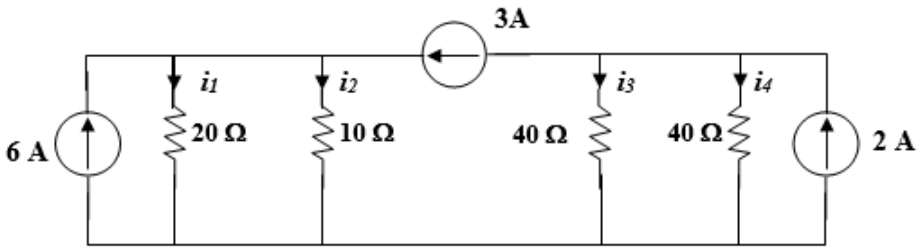
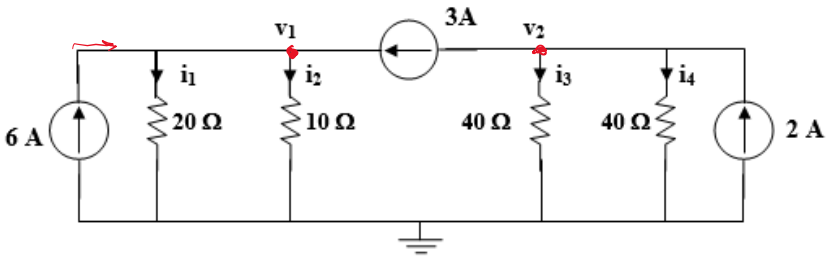


Figure 3.53
For Prob. 3.4.

Chapter 3, Solution 4



At node 1,

$-6 + v_1/(20) + v_1/(10) - 3 = 0$ or $v_1 = 60$ V

At node 2,

$3 - 2 + v_2/(40) + v_2/(40) = 0$ or $v_2 = -20$ V

$$i_1 = v_1/(20) = \mathbf{3\ A},\ i_2 = v_1/(10) = \mathbf{6\ A},$$
$$i_3 = v_2/(40) = \mathbf{-500\ mA},\ i_4 = v_2/(40) = \mathbf{-500\ mA}.$$

Answer: $i_1 = 3$ A, $i_2 = 6$ A, $i_3 = -0.5$ A, $i_4 = -0.5$ A

5. Chapter 3, Problem 5.

Obtain v_0 in the circuit of Fig. 3.54 using Mesh analysis.

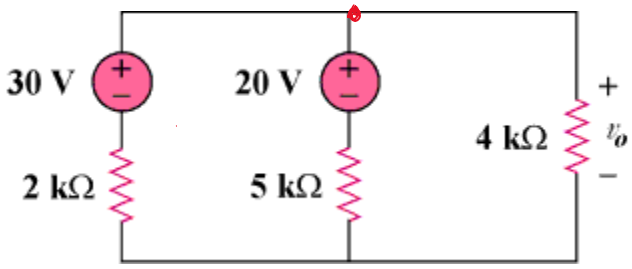


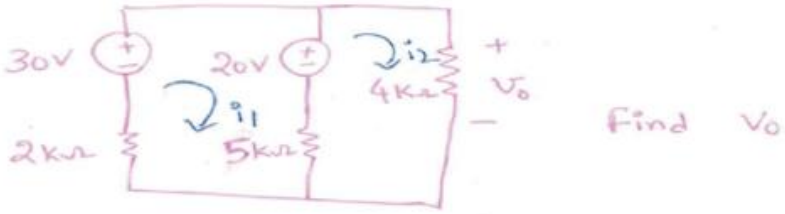
Figure 3.54

For Prob. 3.5.

Chapter 3, Solution 5

Methode 1: with Nodal, Apply KCL to the top node.

$$\frac{30 - v_0}{2k} + \frac{20 - v_0}{5k} = \frac{v_0}{4k} \longrightarrow v_0 = \mathbf{20\ V}$$



Mesh Analysis: Assign clockwise mesh currents i_1 and i_2.

Mesh 1, KVL:

$-30 + 20 + 5(i_1 - i_2) + 2 i_1 = 0$

$7 i_1 - 5 i_2 = 10$ (1)

Mesh 2, KVL:

$-20 + 4 i_2 + 5(i_2 - i_1) = 0$

$-5 i_1 + 9 i_2 = 20$ (2)

Solving (1) and (2): $i_1 = 5$ mA, $i_2 = 5$ mA

$v_o = (4\ k\Omega)i_2 = (4\ k\Omega)(5\ mA)$

Answer: $v_o = 20$ V

6. Chapter 3, Problem 6.

Solve for V_1, in Fig. 3.55, using Mesh analysis.

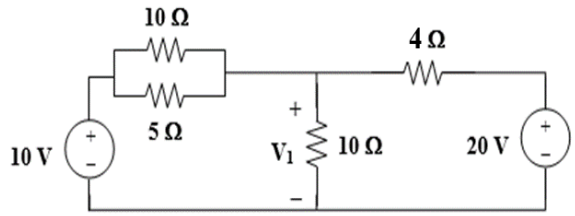
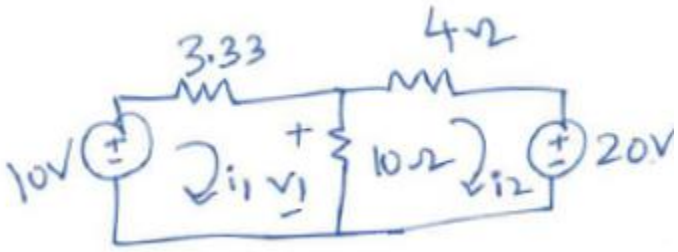


Figure 3.55
For Prob. 3.6.

Chapter 3, Solution 6



First combine the parallel resistors:

$10\ \Omega \parallel 5\ \Omega = (10\times5)/(10+5) = 10/3\ \Omega \approx 3.33\ \Omega$

Assign clockwise mesh currents i_1 and i_2.

Mesh 1:

$-10 + (10/3)i_1 + 10(i_1 - i_2) = 0$

$(40/3)i_1 - 10i_2 = 10 \qquad (1)$

Mesh 2:

$10(i_2 - i_1) + 4i_2 + 20 = 0$

$-10i_1 + 14i_2 = -20 \qquad (2)$

Solving (1) and (2):

$i_1 = -9/13\ A \approx -0.692\ A, \qquad i_2 = -25/13\ A \approx -1.923\ A$

$V_1 = 10(i_1 - i_2) = 10[(-9/13) - (-25/13)]$

Answer: $V_1 = 160/13\ V \approx 12.31\ V$

7. Chapter 3, Problem 37.

Using mesh analysis, find v_0 in the circuit in Fig. 3.57.

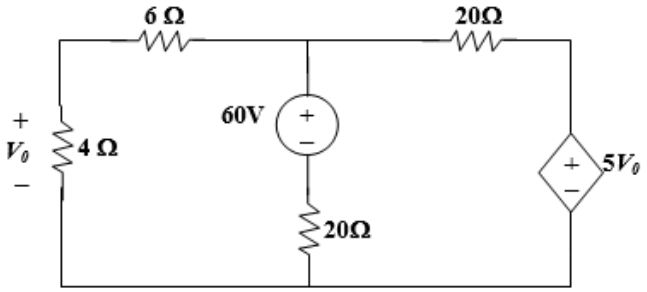
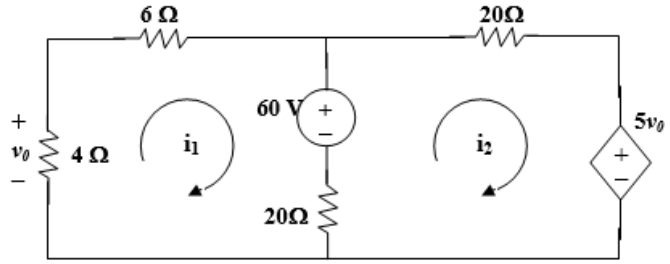


Figure
For Prob. 3.37.

Chapter 3, Solution 37



Applying mesh analysis to loops 1 and 2, we get,

$$4\,i_1+6\,i_1+60+20i_1-20i_2=0 \qquad 30i_1-20i_2+60=0 \qquad \text{which leads to}$$
$$i_2=1.5i_1+3 \qquad (1)$$

$$-20i_1+20i_2-60+20i_2+5v_0=0$$
$$-20i_1+40i_2-60+5v_0=0 \qquad (2)$$

$$\text{But, } v_0=-4i_1 \qquad (3)$$

Using (1), (2), and (3) we get $-20i_1+60i_1+120-60-20i_1=0$ or

$$20i_1=-60 \text{ or } i_1=-3 \text{ amps and } i_2=1.5i_1+3=1.5(-3)+3=-1.5 \text{ amps.}$$

Therefore, we get,

$$v_0=-4i_1=\mathbf{12 \text{ volts}}.$$

8. Chapter 3, Problem 41.

Apply mesh analysis to find ***i*** in Fig. 3.87.

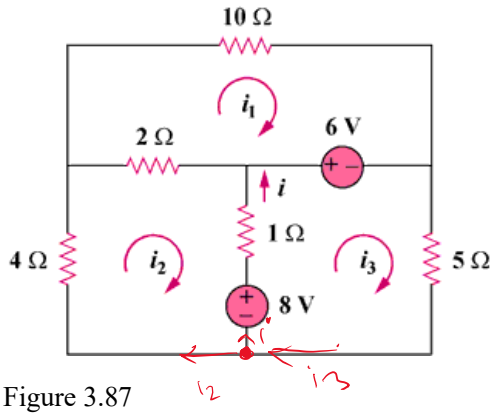


Figure 3.87

For Prob. 3.41.

Chapter 3, Solution 41

For loop 1,

$$6 = 12i_1 - 2i_2 \longrightarrow 3 = 6i_1 - i_2 \qquad (1)$$

For loop 2,

$$-8 = -2i_1 + 7i_2 - i_3 \qquad (2)$$

For loop 3,

$$-8 + 6 + 6i_3 - i_2 = 0 \longrightarrow 2 = -i_2 + 6i_3 \qquad (3)$$

We put (1), (2), and (3) in matrix form,

$$\begin{bmatrix} 6 & -1 & 0 \\ 2 & -7 & 1 \\ 0 & -1 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \\ 2 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 6 & -1 & 0 \\ 2 & -7 & 1 \\ 0 & -1 & 6 \end{vmatrix} = -234, \quad \Delta_2 = \begin{vmatrix} 6 & 3 & 0 \\ 2 & 8 & 1 \\ 0 & 2 & 6 \end{vmatrix} = 240$$

$$\Delta_3 = \begin{vmatrix} 6 & -1 & 3 \\ 2 & -7 & 8 \\ 0 & -1 & 2 \end{vmatrix} = -38$$

At node 0, $i + i_2 = i_3$ or $i = i_3 - i_2 = \dfrac{\Delta_3 - \Delta_2}{\Delta} = \dfrac{-38 - 240}{-234} = \mathbf{1.188\ A}$

9. Chapter 3, Problem 54

Given the circuit Fig. 3.99, use mesh analysis to find the mesh currents.

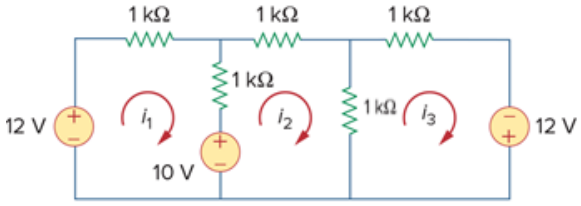


Figure 3.99
For Prob. 3.54

Chapter 3, Solution 54

Let the mesh currents be in mA.

Mesh 1:

$-12 + i_1 + (i_1 - i_2) + 10 = 0$

$2i_1 - i_2 = 2$ (1)

Mesh 2:

$-10 + (i_2 - i_1) + i_2 + (i_2 - i_3) = 0$

$-i_1 + 3i_2 - i_3 = 10$ (2)

Mesh 3:

$(i_3 - i_2) + i_3 - 12 = 0$

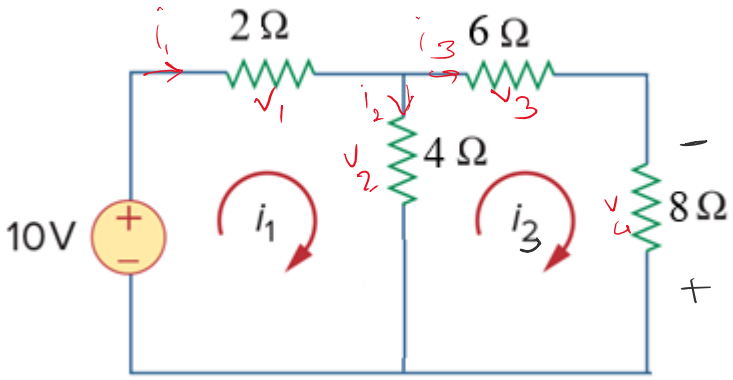
$-i_2 + 2i_3 = 12$ (3)

Solving (1)-(3):

Answer: $I_1 = 5.25$ mA, $I_2 = 8.50$ mA, $I_3 = 10.25$ mA

10. Chapter 3, Problem x

Given the circuit below, use mesh analysis to find the mesh currents voltages v_1, v_2 and v_3.



Assign clockwise mesh currents i_1 and i_3.

Loop 1:

$-10 + 2i_1 + 4(i_1 - i_3) = 0$

$6i_1 - 4i_3 = 10$ (1)

Loop 2:

$4(i_3 - i_1) + 6i_3 + 8i_3 = 0$

$-4i_1 + 18i_3 = 0$ (2)

Solving (1) and (2):

$i_1 = 45/23$ A ≈ 1.957 A

$i_3 = 10/23$ A ≈ 0.435 A

Current through the 4-Ω branch: $i_2 = i_1 - i_3 = 35/23$ A ≈ 1.522 A

$v_1 = 2i_1 = 90/23$ V ≈ 3.913 V

$v_2 = 4i_2 = 140/23$ V ≈ 6.087 V

$v_3 = 6i_3 = 60/23$ V ≈ 2.609 V

Answer: $i_1 \approx 1.957$ A, $i_2 \approx 1.522$ A, $i_3 \approx 0.435$ A; $v_1 \approx 3.913$ V, $v_2 \approx 6.087$ V, $v_3 \approx 2.609$ V